Machine Guardian modelV2 — Review Session Report

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modelV2 works end to end and is verified on the accelerator's RTL, and it caught every bearing tested (7 of 7 development, 4 of 4 locked test) with no healthy false alarms. But on the bearings checked, its network makes the same calls as the simpler anomaly detector it was trained from, so it does not yet add detection ability of its own.

What we did

All work happened on 11 Nov 2026, on the Windows PC (AMD Ryzen 7 3700X, Python 3.12, Vivado 2019.1), not on the ZedBoard.

Step	What we did	Outcome
1	Added five Claude Code agents for the ML work	`.claude/agents/`: model engineer, INT8/hardware checker, diagnostics analyst, experiment auditor, explainer. Not committed
2	Ran modelV2 on this PC and compared it with Run 3	Raw-data replay, bit-exact INT8 check and RTL simulation all passed
3	Reorganised the model documents	Every model document now in `docs/Model/`, numbered 01–13 by date. Staged on `main`, not committed
4	Reviewed modelV2 for improvements	Six recommendations, led by running it on the board
5	Tested the network against the detector alone	On B02–B04 the network makes the same calls as the detector
6	Calculated accuracy, precision, recall, F1 and health-index error	About 99 % row accuracy; health index 4–7 points from its target

modelV2 at a glance

modelV2 is a 1,044-parameter INT8 network that turns one accelerometer's vibration into a bearing status and a health %. It was frozen on 10 Nov and replaces Run 3 (3 Nov), the newest model in `model/`.

	modelV2 (frozen 10 Nov)	Run 3, configuration E (3 Nov)
Network	13 → 32 → 16 → 4, ReLU, INT8	8 → 32 → 16 → 2, ReLU, INT8
Parameters	1,044 (992 weights + 52 biases)	850 (800 + 50)
Inputs	7 vibration z-scores (RMS, peak, kurtosis, crest, 3 frequency bands), the detector's score, 4 averages of the last 4 rows, rpm	6 vibration z-scores, rpm, temperature (always fed 0)
Outputs	3 status logits (healthy / developing / faulty) + a trained health logit	1 deterioration logit + an untrained health head
Host display	Normal / Anomaly / Warning / Failing (latched), health % that never rises, separate temperature warning	Alarm after 3 rows with p > 0.5, latched
Signal	Rear accelerometer, raw 64 or 128 kHz resampled to 42 kHz, low-pass 15 kHz	Same accelerometer, 42.67 kHz features
Window	1.6 s (67,200 samples), one measurement every 12–20 s	1.6 s every ~12 s
Baseline	Each bearing's first 100 measurements; restarts after a vibration step	Each bearing's first 100 measurements
Training labels	Faulty = every row after the detector's sustained onset; health target = a log-scale severity of the detector score	100 healthy + 5 end-stage rows per bearing

	modelV2 (frozen 10 Nov)	Run 3, configuration E (3 Nov)
Training loop	Adam, lr 1e-3 cosine to 0, weight decay 0.01, 4,000 steps, batch 64 (half status, half health), masked two-head loss, seeds 0–2	Adam, lr 1e-3, weight decay 0.01, 2,000 steps, batch 64 (half per class), seeds 0–4
Data	7 Paderborn bearings (B02, B03, B04, B08, B09, B11, B12), 8,943 measurements, leave-one-bearing-out	Same 7 bearings, about 9,000 measurements
Locked test	Run once on B06, B07, B13, B17	Never run
Accelerator time (RTL)	474 cycles = 2.21 µs per measurement	382 cycles = 1.78 µs
B12	Caught, 40.6 min before the end	Missed on all 5 seeds

The B12 catch comes from two changes: training "faulty" on every row after onset rather than only the last 5 extreme rows, and restarting the baseline after a vibration step.

Running modelV2 on this PC

Every check passed: the frozen INT8 model replayed a raw bearing end to end and matched the golden model and the RTL bit for bit.

Check	Result
Live replay of raw bearing B03 (614 `.mat` files) through `live.py`	Finished in 31 s, about 50 ms per measurement. 100 measurements of commissioning, then 447 Normal, 2 Anomaly, and Failing latched at measurement 549, the same row as the saved evaluation. Health 95.4 % while healthy, 79 % at the alarm, 16.4 % at the end. No temperature warning
Golden model vs stored test vectors	Bit-exact: all 64 × 4 sums match `vectors_C.hex`

Check	Result
RTL simulation (`tb_mlp.sv` , Vivado xsim)	Pass, bit-exact. 474 cycles for 1 row (2.21 µs at 214.286 MHz); 1,986 cycles for 64 rows (9.27 µs)
Run 3 model, same checks	Bit-exact and RTL pass, 382 cycles per row
Feature extraction per measurement (resample, filter, FFT)	12.8 ms on the Ryzen
FP32 network in PyTorch	40 µs per row
Python model of the chip's INT8 maths	195 µs per row

The network itself is tiny next to the feature extraction around it. On the ZedBoard's ARM Cortex-A9 the 12.8 ms is likely to grow 10–50×, so the accelerator's 2 µs will be a small part of the time per measurement. Nothing has run on the board yet.

Results: accuracy, precision, recall, F1

modelV2 scores 100 % on the clearly labelled rows and about 99 % on every measurement, but those labels come from the anomaly detector (see Limits). Each bearing was scored by models that never saw it; numbers are the mean of 3 seeds.

Scored against	Accuracy	Precision	Recall	F1
Clearly labelled rows (first 150 healthy + last 20 end-stage per bearing, 1,190 rows), latched alarm	100 %	100 %	100 %	1.00
Same rows, the network's call on each measurement alone	99.6 %	96.6 %	100 %	0.98
Every measurement, against "faulty after the detector's onset"	99.1 %	98.7 %	99.6 %	0.99
Every measurement, latched alarm	98.0 %	96.6 %	99.5 %	0.98

Per bearing (every measurement, seed 0), B03 and B04 score 100 % and B02, B08, B09 and B11 score 99.2–99.9 %. B12 is the weak one: 93 % accuracy, 50 % precision, F1 0.65, because the network called a stretch before the detector's onset faulty.

At the bearing level, the end stage was caught on 7 of 7 development bearings, with 0 healthy measurements alarmed. On the locked test (B06, B07, B13, B17, run once), it was caught on 4 of 4, with 0 of 600 healthy measurements alarmed. INT8 changed 0.06 % of status decisions and no first-Failing row.

Motor Health Index

The health % is within 4–7 points of its target on average and reads about 97 % when healthy, but it never reaches 0 % and is badly off on one test bearing (B06).

Measure	Result
Mean error vs target (B02–B04, all measurements)	4.1–6.6 points
Measurements within 10 points of target (B02–B04)	72–90 %
Correlation with target (B02–B04)	0.995–0.999
Healthy bearing shows	96–97 %
End of life, shown vs target (B02–B04)	12–19 % shown vs 0–1 % target
Mean error before / after onset, 7 development bearings	2.6–9.1 / 5.9–20.4 points (worst: B12)
Mean error before / after onset, locked test	2.9–5.1 / 5.4–51.0 points (worst: B06)
Ottawa locked bearings, mean shown by stage	healthy 97 %, stage 1 16 %, stage 2 13 %

The target is not a measured health. It is a log-scale severity of the anomaly detector's score: 100 % at or below a score of 2.5, 0 % at 25. So "4–7 points of error" means the network copies that formula well, not that it reads true bearing health to within 5 %. It describes how bad the vibration is now, not how long the bearing has left.

Does the network add anything?

On B02–B04, no: the detector alone, with the same "3 rows in a row" rule, fails each bearing at the same measurement as the network, or earlier. Each bearing was scored by the fold models that never saw it; all 3 seeds gave the same rows.

Bearing	Network	Detector > threshold	Detector > 2× threshold	Network without its score input
B02	row 814 (3.1 min after onset)	row 689 (35.1 min before onset)	row 814 (3.1 min after)	row 814 (3.1 min after)
B03	row 548 (0.6 min after)	row 548 (0.6 min after)	row 553 (2.2 min after)	row 551 (1.6 min after)
B04	row 391 (0.6 min after)	row 391 (0.6 min after)	row 392 (0.9 min after)	row 392 (0.9 min after)

Every variant caught the end stage and alarmed 0 healthy measurements. Other evidence:

- The network's faulty score tracks the detector score with a correlation of 0.96–0.99. Row by row, they agree on 89 % (B02), 99.8 % (B03) and 99.2 % (B04) of measurements.
- Removing the detector-score input moves the alarm only 0–3 rows. The tie to the detector comes from the labels: "faulty" was defined as "after the detector's onset".
- The health output tracks the detector score with a correlation of −0.97 to −0.99. Without the score input, its error rises from 4–7 to 6–15 points.

The network is therefore a learned copy of two things the ARM could compute directly. That is still a valid hardware demonstration, but the report should not claim the network detects damage better than the detector. Whether B02's 35 extra minutes are a true early warning is unknown, since the real damage onset is not recorded.

Recommendations

First, run modelV2 on the ZedBoard; then decide whether the network should earn its place by beating the detector.

1. **Run it on the ZedBoard and time the whole measurement.** Measure feature extraction on the ARM plus the accelerator call, not only the 2 µs network. This is the project's headline number and is still simulation only.
2. **Report the detector-only baseline beside the network.** Add it to `modelV2/evaluate.py` and the report, so the network is always judged against it.

3. **Aim for earlier warning.** Try envelope or bearing-defect-frequency features (BPFO/BPFI) and longer trend inputs (slope over 50–100 measurements). Success = alarming earlier than the detector on held-out bearings.
4. **Use the unused development bearings.** B10, B14 and B15 were never used, and B14/B15 cover the constant-speed regime. B05 and B16 could add healthy or unlabelled rows.
5. **Simplify the outputs.** "Developing" cannot be learned (version A never trains it; the Ottawa pre-checks failed). Drop it, or mark it unused in `model.h` and the C code.
6. **Smaller fixes.** Look into the health error on B06 after onset (51 points), tune the status ladder settings (3 to warn, 20 to clear, 3 to fail), and write the real sensor source in `live.py`.

The locked test has now been used. Any change above makes a new model, judged only by leave-one-bearing-out validation unless new test bearings are set aside first.

If the deadline is close: keep modelV2 as is, get it on the board, and present the detector and network honestly as a combined system. With a few more weeks: try the envelope-feature version against the detector-only baseline.

Repository changes

Two sets of changes sit uncommitted in the repository: the model documents moved into one numbered folder, and five project agents.

Model documents. Every model document is now in `docs/Model/`, numbered in the order the work happened. This builds on an unmerged 9 Nov reorganisation from the `MLmodel` branch, extended to today. The changes are staged on `main`.

#	Document	Date
01	Ottawa three-accelerometer model E0–E3	27 Oct
02 (+ 02a–c)	Ottawa single-accelerometer model (+ model, comparison and hardware reports, moved from `model/SINGLE_ACCEL_*.md`)	28 Oct
03	Paderborn B01, first bearing	1 Nov
04	Paderborn anomaly detector baseline	1 Nov
05	Multi-task MLP architecture	1 Nov

#	Document	Date
06	Dataset validation and training plan	2 Nov
07 (+ 07a–e)	B02–B04 processing and labels (+ label review records)	2 Nov
08	Training run 1	3 Nov
09	Training run 2	3 Nov
10	Training run 3	3 Nov
11	Development history, Ottawa to Run 3	3 Nov
12	Run 3 model summary (was `00_Current_model`)	9 Nov
13	modelV2 bearing status and health (current model)	10 Nov

The highest number is always the current model, and new documents take the next number. `docs/Model/README.md` gives the reading order and maps old file names to new ones. The report builders, `modelV2/report.py`, and the root, `model` and `modelV2` READMEs all point at the new names.

Project agents (`.claude/agents/`, untracked):

- `ml-model-engineer`: builds, trains and fixes models under the project's experiment rules.
- `int8-hardware-checker`: INT8 export, golden model and RTL bit-exactness.
- `model-diagnostics-analyst`: finds out why the model behaves as it does, writing only to scratch.
- `ml-experiment-auditor`: read-only review for leakage, test-set peeking and report numbers.
- `model-explainer`: read-only explanations grounded in the docs and code.

Limits

The results are honest but rest on few bearings and on labels drawn from the detector, so treat the high scores with care.

- **Few bearings.** 7 development and 4 locked test bearings. With so few, the true catch rate could plausibly be anywhere from about 60 % to 100 %.

- **Scores are high partly by construction.** The healthy rows are the same rows each bearing's baseline is built from, and the end-stage rows are extreme. A simple "is vibration way up?" rule also scores near 100 % on them.
- **Labels and the health target come from the detector.** The 99 % accuracy largely measures agreement with the detector, and the health target is a formula, not measured health. True damage onset and remaining life are not known for any bearing.
- **The detector comparison covered only B02–B04**, the bearings with raw data on this PC. Removing the score input was done at inference, not by retraining.
- **Locked test saved only at the bearing level.** Row-level accuracy is not available for it.
- **Not run on the ZedBoard.** All timing and correctness results are from this PC and RTL simulation.